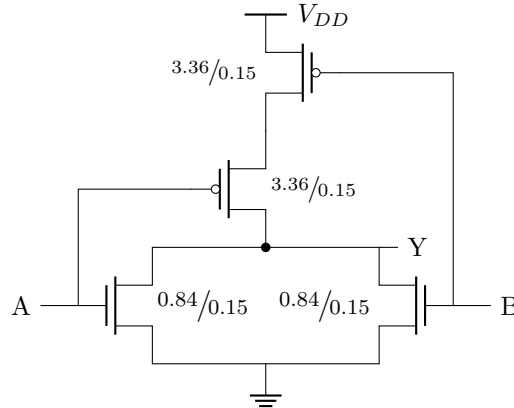
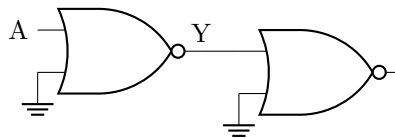


EE5311 Digital IC Design: Assignment 7
NOR gate – schematic and layout extracted simulation

1. Construct a two-input NOR gate shown below. Use appropriate number of fingers for all the devices considering the layout constraints.



The NOR gate drives another NOR gate of same size as shown below:



The input A is a pulse between 0 and V_{DD} , with a rise and fall time of 5 ps and a pulse width of 400 ps. Input B is connected to ground.

- (a) Obtain the delay of the NOR gate for $V_{DD} = 1.8\text{V}$. How does this delay compare against the one calculated using the logical effort expression?
- (b) Draw the layout containing two NOR gates connected as shown above and measure the layout extracted delay for $V_{DD} = 1.8\text{V}$.